

Projectile Motion

Core Concepts & Equations

- A two-dimensional motion under uniform gravity
- Independent treatment of horizontal and vertical motion
- Trajectory, range, maximum height, and time of flight
- Simple theory explained with clear diagrams

1. What Is Projectile Motion?

Projectile motion describes the curved path traced by an object that is launched into the air and then moves freely under the influence of gravity alone. Once launched, the only force acting on the object is its weight, so the motion is entirely predictable using the equations of uniformly accelerated motion. Common examples include a kicked football, a thrown ball, a bullet fired from a gun, or water arcing out of a fountain.

The defining feature of projectile motion is that it can be split into two completely independent components: horizontal motion, which happens at constant velocity, and vertical motion, which happens under constant downward acceleration due to gravity. Treating these two directions separately is the single most important idea in this topic.

1.1 Assumptions Used in This Model

- Air resistance is neglected.
- The acceleration due to gravity, g , is constant ($\approx 9.8 \text{ m/s}^2$, downward).
- The Earth's curvature and rotation are ignored (flat-ground approximation).
- The object is treated as a point mass, so spin and shape effects are ignored.

1.2 Trajectory Shape

Because horizontal displacement grows linearly with time while vertical displacement involves a time-squared term, eliminating time between the two equations shows that the path traced out is a parabola.

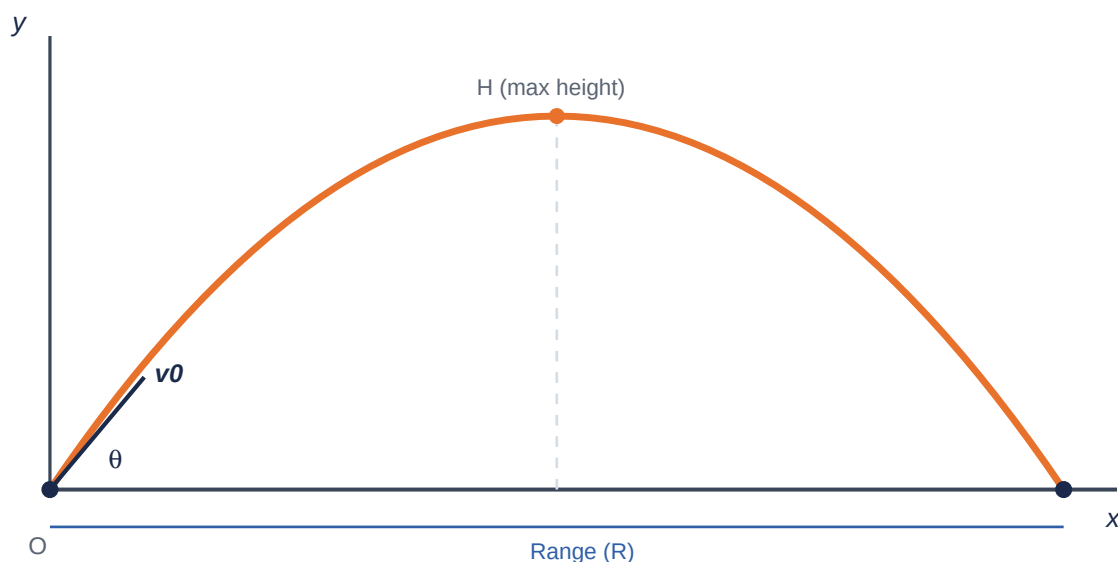


Figure 1 — A projectile launched at angle θ with initial speed v_0 follows a parabolic path.

2. Resolving the Initial Velocity

When a projectile is launched with initial speed v_0 at an angle θ above the horizontal, that velocity is first split into horizontal and vertical components using basic trigonometry.

$$v_x = v_0 \cos \theta \quad v_y = v_0 \sin \theta$$

2.1 Horizontal Motion (constant velocity)

There is no horizontal force acting on the projectile (air resistance is ignored), so the horizontal component of velocity never changes throughout the flight.

$$x = v_x t = (v_0 \cos \theta) t$$

2.2 Vertical Motion (constant acceleration)

In the vertical direction, gravity continuously decelerates the projectile on the way up and accelerates it on the way down. The standard equations of motion under constant acceleration apply, with $a = -g$.

$$v_y = v_0 \sin \theta - g t$$

$$y = (v_0 \sin \theta) t - \frac{1}{2} g t^2$$

These two independent sets of equations — one for x , one for y — are all that is needed to fully describe the position and velocity of the projectile at any instant in time.

3. Key Results: Time, Height and Range

Combining the horizontal and vertical equations produces three results that are used constantly when solving projectile problems. They assume launch and landing occur at the same vertical height.

Quantity	Formula	Meaning
Time of flight (T)	$T = 2 v_0 \sin \theta / g$	Total time in the air
Maximum height (H)	$H = v_0^2 \sin^2 \theta / (2g)$	Peak height above launch point
Horizontal range (R)	$R = v_0^2 \sin(2\theta) / g$	Horizontal distance travelled

3.1 Maximum Range

Since R depends on $\sin(2\theta)$, and sine reaches its maximum value of 1 when its argument is 90° , the range is greatest when $2\theta = 90^\circ$, i.e. when the launch angle $\theta = 45^\circ$. Angles that differ equally from 45° in either direction (for example 30° and 60°) produce the same range.

3.2 Equation of the Trajectory

Eliminating time t from the x and y equations gives the trajectory directly as a function of horizontal position — confirming the parabolic shape seen in Figure 1.

$$y = x \tan \theta - g x^2 / (2 v_0^2 \cos^2 \theta)$$

4. Nature of the Motion

4.1 Symmetry About the Peak

When a projectile lands at the same height from which it was launched, its path is symmetric about the highest point. The time taken to rise to maximum height equals the time taken to fall back down, so each half of the flight lasts exactly $T/2$. The speed at any given height is the same on the way up as it is on the way down, though the vertical component of velocity points in opposite directions at those two instants.

4.2 Velocity Along the Path

The horizontal velocity component stays fixed throughout the flight, while the vertical component steadily decreases in magnitude as the projectile rises, becomes zero exactly at the peak, and then grows again in the downward direction as it falls. At every point on the trajectory, the resultant velocity is directed along the tangent to the parabolic path, which is why the projectile appears to travel fastest at launch and landing and slowest near the top.

4.3 What Determines the Shape of the Path

Three quantities control the entire trajectory: the initial speed, the launch angle, and the strength of gravity. A larger initial speed stretches the parabola wider and higher for a given angle. Increasing the launch angle (up to 45°) increases the height gained but does not always increase the range, since range depends on the balance between height and horizontal speed. A stronger gravitational field pulls the projectile down more quickly, producing a shorter, flatter path for the same launch conditions.

5. Real-World Applications

Sports — Analysing the flight of a football, basketball, javelin, or golf ball to optimise launch angle and power for maximum distance or accuracy.

Ballistics — Predicting the trajectory of bullets, artillery shells, and rockets, including corrections for range and drop over long distances.

Engineering — Designing water fountains, firefighting hoses, and irrigation sprinklers where the landing point of a fluid stream must be controlled.

Space Science — Modelling the free-fall arcs of objects released from aircraft or spacecraft, and as a stepping stone to understanding orbital mechanics.

5.1 Quick Revision Summary

Concept	Key Idea
Independence of motion	Horizontal and vertical motions occur independently
Horizontal motion	Constant velocity (no acceleration)
Vertical motion	Constant acceleration g , directed downward
Path shape	Parabola
Maximum range angle	45° (equal launch/landing height)

End of Notes